

Creative Immersive AI: Emerging Challenges and Opportunities for Creative Applications of AI in Immersive Media

Bektur Ryskeldiev
University of Tsukuba, Mercari R4D
Japan
bektour@digitalnature.slis.tsukuba.ac.jp

Suzana Ilić
MLT Labs
Japan
suzana@mlt.ai

Yoichi Ochiai
University of Tsukuba
Japan
wizard@slis.tsukuba.ac.jp

Luba Elliott
Elluba.com
United Kingdom
luba.elliott@gmail.com

Helena Nikonole
ITMO University
Russian Federation
helenanikonole@gmail.com

Mark Billingham
University of Auckland
New Zealand
mark.billinghurst@auckland.ac.nz

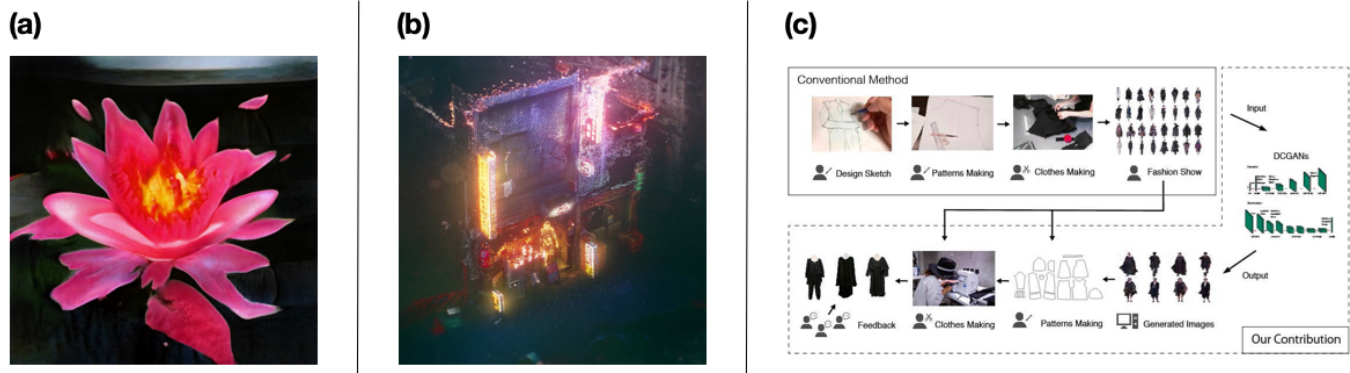


Figure 1: Examples of computational art: (a) an image generated via NVIDIA StyleGAN 2 model, using photographs of flowers and paintings of Francis Bacon as an input dataset [20], (b) an immersive scene made by creating a scan of a physical space represented as 3D point clouds [7]. (c) Example of a system that aids in creative fashion clothing design process using deep learning [11]

ABSTRACT

Over the past several years artificial intelligence (AI) techniques have gained a considerable presence in immersive media. From assistance with real-time digital production to emerging novel forms of creative expression, applications of AI are becoming ubiquitous in the fields of human-computer interaction (HCI), computer graphics, and media art. As we are interested in how novel computational techniques are shaping the state of creativity in immersive and interactive technologies, we organize this special interest group (SIG) to stimulate a discussion among AI, HCI, immersive media, and art communities. The goal of this SIG includes outlining existing and emerging areas of cross-disciplinary collaborations, proposing a roadmap of future goals and challenges for creative immersive

AI research, and establishing a diverse group of researchers and practitioners involved in creative applications of AI in immersive and interactive media.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → **Computer graphics**; **Artificial intelligence**; **Machine learning**; **Mixed / augmented reality**; **Virtual reality**; • **Applied computing** → **Media arts**.

KEYWORDS

artificial intelligence, augmented reality, virtual reality, mixed reality, extended reality, human-computer interaction, immersive media, media art, computer graphics

ACM Reference Format:

Bektur Ryskeldiev, Suzana Ilić, Yoichi Ochiai, Luba Elliott, Helena Nikonole, and Mark Billingham. 2021. Creative Immersive AI: Emerging Challenges and Opportunities for Creative Applications of AI in Immersive Media. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8095-9/21/05.

<https://doi.org/10.1145/3411763.3450399>

(CHI '21 Extended Abstracts), May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3411763.3450399>

1 INTRODUCTION

Artificial intelligence (AI) methods and tools are becoming increasingly common in creative applications of immersive and interactive media. Amato et al. [3] divide them into three categories: *creation*, or the development of new AI techniques for artistic expression, *production*, or tools and systems where AI is being used, and *consumption*, or diffusion of the former two groups into final immersive AI artworks and installations.

Recent interest towards novel creative computational techniques can be attributed to such themes as exploration of human versus robot perception (e.g., “DeepDream” [17, 24]), creating synthetic media by replacing certain elements of existing videos (“Deepfakes” [21, 22]), or generating completely new, “never-before-seen” artificial images based on existing datasets (generative adversarial networks (or GANs) [10, 25]).

Furthermore, recent applications of machine learning have shown considerable results in such computer vision and human-computer interaction (HCI) problems as drawing assistance through autocompletion [8], object detection [18], pose estimation [12], image segmentation [5], and object and scene reconstruction [9] – most of which are used to build basic features and interactions offered by modern augmented and virtual reality frameworks, used both in mobile and desktop applications (e.g., ARKit¹ or Vuforia²).

With an ongoing impact in both visual arts and immersive media, AI is attracting attention among artists, researchers, and industry practitioners. Jun Rekimoto [19] projects that in the near future AI would become and integral part of future interfaces, with immersive displays being one of the primary form of communication between humans and artificial entities. Similarly, the effects of AI on creativity are actively discussed in studies [2, 11], and workshops on human augmentation [14], intelligent tools in creative processes [13], and on machine learning for creativity and design [6].

In parallel to academia, a public interest for computational creativity tools is represented in part by such projects as RunwayML³ and Magenta⁴ offering a simplified method of working with model training and output, as well as by such solutions as CoreML⁵, Fritz AI⁶, or Azure Cloud⁷ offering a cross-platform approach to hosting same or similar models across multiple devices and systems.

At the same time, there is also a demand for consumption of immersive AI art. For instance, recent notable immersive and mixed media artworks using AI techniques as the main form of artistic expression include installations by Refik Anadol [4] and Memo Akten [1], as well as multiple permanent exhibits at The National Museum of Emerging Science and Innovation of Japan (Miraikan) [15, 16] – exploring the topics of memory, perception, and human-AI coexistence through applications of deep learning.

¹<https://developer.apple.com/augmented-reality/>

²<https://developer.vuforia.com/>

³<https://runwayml.com/>

⁴<https://magenta.tensorflow.org/>

⁵<https://developer.apple.com/machine-learning/>

⁶<https://www.fritz.ai/>

⁷<https://azure.microsoft.com/en-us/overview/ai-platform/>

2 THEMES

Given the broad scope of creative applications of AI in immersive media, we outline several main themes that will be discussed at SIG meeting:

- **Art** – in particular AI artwork made with GANs, generative music and design, as well as new and mixed media art.
- **Augmented, Virtual, and Extended Reality (AR/VR/XR)** – exploring ideas for creative immersive applications that work in conjunction with machine learning, such as AR copy-paste (e.g., ClipDrop⁸), optimized capture of scenes and objects through point clouds, and creative real-time AR filters (e.g., Meisai⁹).
- **Tools**, in particular a discussion on software components, applications, and systems that empower creativity through combinations of AI and immersive media.

3 GOALS

Based on these themes we aim to lead the discussion towards achieving the following goals:

- **Cross-disciplinary collaborations.** We aim to discuss and outline potential vectors for cross-disciplinary collaborations among artists, researchers, and practitioners. Examples of such include, but are not limited to, applying novel methods of immersive media generation and display to mixed media art and building community-driven tools for creativity and design.
- **Future goals and challenges.** In parallel to a discussion on collaborations, we also expect to create a roadmap of potential future goals and challenges for creative immersive AI. This plan will include such topics as, for instance, guidelines on ethical usage of datasets in application to open-world 3D photogrammetry (as there are already privacy concerns in regards to outdoor scanning in AR/VR applications [23]).
- **Building a community.** Finally, through this SIG meeting we aim to create a community of artists, researchers, and practitioners interested in creative applications of AI. We aim to attract a diverse group of attendees from the international machine learning communities, as well as specialized groups such as SIGGRAPH, SIGCHI, and Ars Electronica, establishing a multi-directional exchange of knowledge.

4 SESSION PLAN

4.1 Before the meeting

Before the SIG meeting we will invite participants through social media promotion and professional mailing lists. We expect attendees of diverse backgrounds with different levels of familiarity with the topic. The attendees will be recruited from international machine learning communities as well as special interest groups. Furthermore, closer to the meeting we plan to create a Slack channel and a shared Google Doc for easier note sharing and networking online.

⁸<https://clipdrop.co/>

⁹<https://kitasenjudeisign.com/meisai/>

4.2 Schedule

Due to the conference format this meeting is intended to be an online-only event, about 75 minutes long, with program consisting of following activities:

- **5 minutes** Introduction of the SIG session, its format and panelists
- **30 minutes** “Creative Immersive AI” – a moderated discussion panel, with participants consisting of invited notable artists, scientists, and industry practitioners, moderated by one of the SIG meeting organizers
- **5 minutes** Preparing for group discussion, attendees are divided into 3-5 groups along with panelists depending on the topics (Art, AR/VR, Tools)
- **30 minutes** Group discussions with the panelists
- **5 minutes** Session closing that summarizes group discussions and proposes the next steps for the SIG meeting

5 POST-MEETING PLANS

During the meeting will collect workshop outputs such as notes, session recordings, and presentations and upload them online for sharing among the session attendees and their professional networks. Based on this material we also plan to write a blog article about SIG meeting, as well as submitting it for publication in media arts magazine (such as Neural¹⁰). After the conference we plan to continue developing this SIG community and organize a workshop and a Birds of a Feather (BoF) panel at CHI and SIGGRAPH conferences respectively.

ACKNOWLEDGMENTS

This work was supported by JSPS KAKENHI Grant Number 19K20313.

REFERENCES

- [1] Memo Akten. 2018. Deep Meditations: A brief history of almost everything. Article. Retrieved December 2, 2020 from <http://www.memo.tv/works/deep-meditations/>.
- [2] Nadisha-Marie Aliman and Leon Kester. 2020. Artificial creativity augmentation. In *Int. Conf. on Artificial General Intelligence*. Springer, 23–33.
- [3] Giuseppe Amato, Malte Behrmann, Frédéric Bimbot, Baptiste Caramiaux, Fabrizio Falchi, Ander Garcia, Joost Geurts, Jaume Gibert, Guillaume Gravier, Hadmut Holken, et al. 2019. AI in the media and creative industries. *arXiv preprint arXiv:1905.04175* (2019).
- [4] Refik Anadol. 2015. DeepDream - a code example for visualizing Neural Networks. Article. Retrieved December 2, 2020 from <https://ai.googleblog.com/2015/07/deepdream-code-example-for-visualizing.html>.
- [5] Liang-Chieh Chen, George Papandreou, Florian Schroff, and Hartwig Adam. 2017. Rethinking Atrous Convolution for Semantic Image Segmentation. *CoRR abs/1706.05587* (2017). [arXiv:1706.05587](https://arxiv.org/abs/1706.05587)
- [6] Luba Eliott, Sander Dieleman, Adam Roberts, Tom White, Holly Grimm, Mattie Tesfaldet, Samaneh Azadi, and Daphne Ippolito. 2020. Machine Learning for Creativity and Design. Article. Retrieved December 4, 2020 from <https://neurips2020creativity.github.io/>.
- [7] Ruben Frosali. 2020. Diorama #1. Video. Retrieved December 2, 2020 from <https://vimeo.com/422714783>.
- [8] David Ha. 2017. Teaching Machines to Draw. Article. Retrieved December 4, 2020 from <https://ai.googleblog.com/2017/04/teaching-machines-to-draw.html>.
- [9] Xian-Feng Han, Hamid Laga, and Mohammed Bennamoun. 2019. Image-based 3D Object Reconstruction: State-of-the-Art and Trends in the Deep Learning Era. *CoRR abs/1906.06543* (2019). [arXiv:1906.06543](https://arxiv.org/abs/1906.06543)
- [10] Tero Karras, Samuli Laine, and Timo Aila. 2019. A style-based generator architecture for generative adversarial networks. In *Proc. of the IEEE Conf. on Computer Vision and Pattern Recognition*. 4401–4410.
- [11] Natsumi Kato, Hiroyuki Ozone, Daitetsu Sato, Naoya Muramatsu, and Yoichi Ochiai. 2018. Deepwear: a case study of collaborative design between human and artificial intelligence. In *Proceedings of the Twelfth International Conference on Tangible, Embedded, and Embodied Interaction*. 529–536.
- [12] Alex Kendall, Matthew Grimes, and Roberto Cipolla. 2015. Convolutional networks for real-time 6-DOF camera relocalization. *CoRR abs/1505.07427* (2015). [arXiv:1505.07427](https://arxiv.org/abs/1505.07427)
- [13] Janin Koch, Jennifer Pearson, Andrés Lucero, Miriam Sturdee, Wendy E. Mackay, Makayla Lewis, and Simon Robinson. 2020. Where Art Meets Technology: Integrating Tangible and Intelligent Tools in Creative Processes. In *Extended Abstracts of the 2020 CHI Conf. on Human Factors in Computing Systems (CHI EA '20)*. Association for Computing Machinery, New York, USA, 1–7. <https://doi.org/10.1145/3334480.3375172>
- [14] Hideki Koike, Jun Rekimoto, Junichi Ushiba, and Asa Furuya, Shinichi and Ito. 2019. The First IEEE VR Workshop on Human Augmentation and Its Applications (HAA2019). Website. Retrieved December 2, 2020 from <https://sites.google.com/view/haa2019>.
- [15] Miraikan. 2019. Digitally Natural - Naturally Digital. Website. Retrieved December 2, 2020 from <https://www.miraikan.jst.go.jp/en/exhibitions/future/digitallynatural/>.
- [16] Miraikan. 2019. The tearoom going from zero to one. Website. Retrieved December 2, 2020 from <https://www.miraikan.jst.go.jp/en/exhibitions/future/zeroone/>.
- [17] Alexander Mordvintsev, Christopher Olah, and Mike Tyka. 2015. DeepDream - a code example for visualizing Neural Networks. Article. Retrieved December 2, 2020 from <https://ai.googleblog.com/2015/07/deepdream-code-example-for-visualizing.html>.
- [18] Joseph Redmon, Santosh Kumar Divvala, Ross B. Girshick, and Ali Farhadi. 2015. You Only Look Once: Unified, Real-Time Object Detection. *CoRR abs/1506.02640* (2015). [arXiv:1506.02640](https://arxiv.org/abs/1506.02640)
- [19] Jun Rekimoto. 2019. Homo Cyberneticus: The Era of Human-AI Integration. *CoRR abs/1911.02637* (2019). [arXiv:1911.02637](https://arxiv.org/abs/1911.02637) <http://arxiv.org/abs/1911.02637>
- [20] Bektur Ryskeldiev. 2020. Bacon Flowers: A Generative Exploration of Metamorphosis Between Human and Nature. Video. Retrieved December 2, 2020 from <https://vimeo.com/479689108>.
- [21] Supasorn Suwajanakorn, Steven M Seitz, and Ira Kemelmacher-Shlizerman. 2017. Synthesizing obama: learning lip sync from audio. *ACM Transactions on Graphics (TOG)* 36, 4 (2017), 1–13.
- [22] Justus Thies, Michael Zollhofer, Marc Stamminger, Christian Theobalt, and Matthias Nießner. 2016. Face2face: Real-time face capture and reenactment of rgb videos. In *Proc. of the IEEE Conf. on Computer Vision and Pattern Recognition*. 2387–2395.
- [23] Jan-Erik Vinje. 2018. Privacy Manifesto for AR Cloud Solutions. Article. Retrieved December 2, 2020 from <https://medium.com/openarcloud/privacy-manifesto-for-ar-cloud-solutions-9507543f50b6>.
- [24] Jason Yosinski, Jeff Clune, Anh Nguyen, Thomas Fuchs, and Hod Lipson. 2015. Understanding neural networks through deep visualization. *arXiv preprint arXiv:1506.06579* (2015).
- [25] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A Efros. 2017. Unpaired image-to-image translation using cycle-consistent adversarial networks. In *Proc. of the IEEE Int. Conf. on Computer Vision*. 2223–2232.

¹⁰<http://neural.it/>